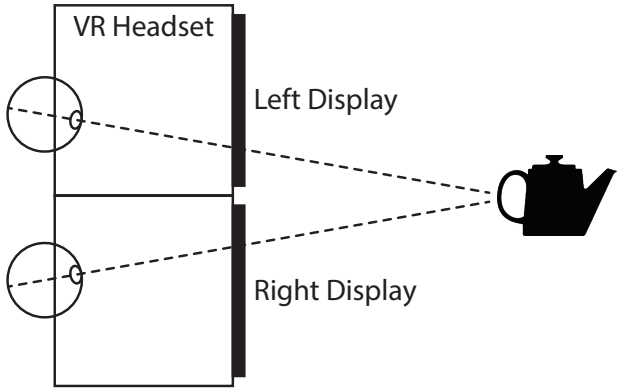
VR, Image Sensors, Image Processing 14

CS 184: Foundations of Computer Graphics

1 Virtual Reality Ethan

Virtual Reality Ethan wears a virtual reality (VR) headset with two displays, one for each eye. A top-down illustration is shown below. Ethan is currently looking at a virtual teapot mesh.



1. In VR, **vergence** refers to the rotation of the eyes to focus on near or far objects. Ethan looks at a teapot that is 1m away in virtual space. Why do his eyes need to rotate (or verge) to adjust for the perceived depth?

2. Now the teapot is located 100m away (effectively at infinity). If Ethan looks at the teapot, what is the angle between Ethan's eyes?

3. **Accommodation** is the eye's ability to change the shape of its lens to bring objects into focus. Why do Ethan's eyes need to accommodate to the VR screens? In the diagram above where the teapot is 1m away, circle the point on which each eye must focus.

4. Why might Ethan experience discomfort while looking at the teapot? Briefly describe the effect that Ethan is experiencing.

5. Ethan quickly turns his head while wearing the headset. Why is *low-latency* motion tracking essential for a realistic and comfortable VR experience?

6. Ethan decides to watch a monoscopic 360° YouTube video that was captured with a single 360° camera. Will there be any perceptual difference between watching this on a *two-display* headset versus a *single-display* headset?

7. Ethan notices that his peripheral vision seems limited in VR. How does field-of-view (FOV) influence immersion in a VR headset, and what are the trade-offs of increasing it?

8. Ethan's headset offers a passthrough mode that shows a live video feed of the real world with virtual objects composited on top. What is this type of system called, and how does it differ from traditional VR?

2 Sensors

The Poisson distribution is a discrete probability distribution used to model the probability of a given number of events occurring in a fixed time interval (under some assumptions).

1. Random variable $X \sim \text{Poisson}(\lambda)$, where $\lambda > 0$, is defined by its probability mass function

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

   What is $\mathbb{E}[X]$?

2. Jennifer has a perfect image sensor, which counts the number N of photon arrivals during a shutter duration, and outputs a pixel value $P = \alpha N$ for some constant $\alpha > 0$. It is known that $N \sim \text{Poisson}(\lambda)$, and $\text{Var}(N) = \lambda$.

   What is the SNR of the pixel output P?

3. Jennifer places this image sensor inside her camera. The photon arrival rate λ is proportional to aperture area. If she wants to increase her SNR by a factor of 2, how should she change the size of the aperture?

3 Poisson Shot Noise

In the following, assume that all the image sensors considered have 100% quantum efficiency and zero read noise, so the only noise present in the signal is due to Poisson shot noise. Each of the following cameras is used to take a picture of a patch of the same, uniformly illuminated white wall, with the same shutter duration across all cameras. The pixels do not saturate in any camera.

Camera 1 has a 10x10mm sensor with 50mm f/2.0 lens and 4 million pixels. The individual pixels have an SNR of 100.

1. Camera 2: 10x10mm sensor with 50mm f/1.0 lens and 4 million pixels.

   SNR = _____

2. Camera 3: 10x10mm sensor with 50mm f/2.0 lens and 1 million pixels.

   SNR = _____

3. Camera 4: 20x20mm sensor with 50mm f/4.0 lens and 16 million pixels.

   SNR = _____

4. Camera 5: 5x5mm sensor with 25mm f/1.0 lens. The individual pixels have an SNR of $100\sqrt{2}$. How many pixels does Camera 5 have?

   Number of pixels = _____